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Evaluating the follicular output rate, follicle-to-oocyte index, and ovarian sensitivity index for bovine application as key performance indicators of ovarian response to follicle-stimulating hormone superstimulation in Holstein cows

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ABSTRACT

Variability in ovarian-stimulation (OS) response limits the efficiency of ovum pick-up (OPU)–in vitro production programs, with a minority of donors often producing most embryos. Reliable indicators to predict OS response in individual cows remain limited. In this study, we evaluated and quantified the ability of 3 key performance indicators adapted for bovine application—follicular output rate (FORT_bov), follicle-to-oocyte index (FOI_bov), and ovarian sensitivity index (OSI_bov)—to classify ovarian response in dairy cows. Records from 1,090 follicle-stimulating hormone–stimulated OPU sessions in 481 Holstein cows were analyzed, accounting for repeated sessions within cow in inference. Using the joint distribution of aspirated follicles and recovered cumulus–oocyte complexes (COC), sessions were classified as poor, normal, or high responders (overall mean ± 1 SD), representing 15.6%, 62.8%, and 21.7% of sessions, respectively. The FOI_bov, FORT_bov, and OSI_bov increased progressively across response groups and were associated with OPU outcomes; OSI_bov showed the strongest correlations with aspirated follicles, total COC, and grade A+B COC. In receiver operating characteristic analyses, OSI_bov discriminated poor from normal + high responders (area under the receiver operating characteristic curve [AUC] = 0.974, 95% CI 0.962–0.983) and high from poor + normal responders (AUC = 0.951, 95% CI 0.934–0.966), exceeding the performance of FOI_bov and FORT_bov; the corresponding optimal OSI_bov cutoffs were 11 and 25, respectively. OSI_bov-based categories agreed well with the clinical classification (quadratic weighted κ_w = 0.777). In single-predictor ordinal logistic regression, OSI_bov yielded the lowest Akaike information criterion and the highest concordance index (C-index = 0.956). Across cows with repeated sessions, antral follicle count and yield traits were highly repeatable, whereas FOI_bov and FORT_bov showed greater within-cow variability. We conclude that OSI_bov is a practical indicator of bovine OS response and may support donor ranking and individualized stimulation to improve embryo-production efficiency.

Key words: ovarian stimulation, follicular output rate, follicle-to-oocyte index, ovarian sensitivity index, OPU-IVP

INTRODUCTION

Since the first embryo transfer (**ET**) derived calf was reported in 1951 (Willett et al., 1951), the ET industry has advanced substantially. In recent years, ovum pick-up (**OPU**) coupled with in vitro embryo production (**IVP**) has become a principal biotechnology to accelerate dissemination of elite genetics by shortening the generation interval and increasing herd-level genetic gain, with clear potential for large-scale commercial application (Monteiro et al., 2017, Ferré et al., 2020, Hayden et al., 2023, Xiao et al., 2025). Within this framework, multiple groups have refined donor management (Leroy et al., 2014; Velazquez, 2023), monitoring of follicular dynamics (Karami Shabankareh et al., 2015; Fry, 2020; Hayden et al., 2022), and laboratory workflows (Huang et al., 2013; Sakaguchi, 2025) to increase the recovery of developmentally competent cumulus–oocyte complexes (**COC**) and improve embryo yield. Nevertheless, overall program efficiency remains constrained by pronounced variability in individual donor responses to ovarian stimulation (**OS**) and OPU.

Heterogeneity among donors is a defining feature of OPU-IVP systems. Even under uniform management and comparable protocols, ovarian response to exogenous gonadotropins varies widely (De Roover et al., 2005; Durocher et al., 2006; Garcia et al., 2020). A small subset of donors

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

Table 1. Definition of clinical ovarian response categories (based on follicles and COC numbers)

Category	Defining criteria
Follicle: Low outcome	<13 aspirated follicles
Follicle: Medium outcome	13 to 39 aspirated follicles
Follicle: High outcome	>39 aspirated follicles
COC: Low outcome	<6 recovered COC
COC: Medium outcome	6 to 22 recovered COC
COC: High outcome	>22 recovered COC
Classification of ovarian response categories	
Poor	Sessions with low follicular outcome combined with low or medium COC outcome, and those with low COC outcome combined with low, medium, or high follicular outcome
Normal	Sessions that did not meet the criteria for either poor or high response
High	Sessions with high follicular outcome combined with medium or high COC outcome, and those with high COC outcome combined with medium or high follicular outcome

contributes disproportionately to total embryo output—approximately 30% of cows may produce 70% of embryos (Durocher et al., 2006; Souza et al., 2015; Cremonesi et al., 2020)—whereas about 24% produce none (Singh et al., 2004). Similar variability has been documented in a cohort of 987 cows (Lerner et al., 1986), illustrating the difficulty of forecasting individual response and complicating the design of cost-effective stimulation strategies.

To more precisely define ovarian stimulation (**OS**) response, several stratification approaches have been proposed. Cushman et al. (1999) categorized animals as low, medium, or high responders based on antral follicle count (**AFC**). Singh et al. (2004) suggested a dynamic metric relating the number of follicles ≥5 mm after superstimulation to the number of follicles ≥2 mm at wave emergence. Other studies combined follicle counts with COC to disentangle technical effects from intrinsic biological variation (De Roover et al., 2005; Durocher et al., 2006). Reserve biomarkers such as circulating anti-Müllerian hormone (**AMH**; Rico et al., 2009; Souza et al., 2015) and AFC (Burns et al., 2005; Mossa and Evans, 2023) show population-level associations with superovulatory outcomes in cattle; however, as static indicators, they provide only a partial view (Cesarano et al., 2022). They do not capture the dynamic efficiency with which a given protocol converts the small antral follicle pool into aspiratable follicles and recoverable oocytes—a process influenced by follicular-wave status, gonadotropin-receptor sensitivity, and protocol design (La Marca et al., 2013). Consequently, although reserve markers correlate with response in large cohorts, their individual-level predictive performance is modest and often leads to misclassification (De Roover et al., 2005; Huber et al., 2013; Yildiz et al., 2024); individuals with apparently poor reserve may respond well, and vice versa.

In human assisted reproduction (**ART**), this limitation has been addressed by process-oriented key performance indicators (**KPI**) that extend beyond simple oocyte counts: the follicular output rate (**FORT**; Gallot et al., 2012; Bessow et al., 2019; Grynberg and Labrosse, 2019), the follicle-to-oocyte index (**FOI**; Alviggi et al., 2018; Carosso et al., 2022; Sunkara et al., 2025), and the ovarian sensitivity index (**OSI**; Huber et al., 2013; Camargo-Mattos et al., 2020; Weghofer et al., 2020). These indices quantify, respectively, (1) the proportion of baseline small antral follicles that reach a preovulatory size under stimulation, (2) COC recovered relative to baseline AFC, and (3) COC per unit of exogenous follicle-stimulating hormone (**FSH**)—thereby emphasizing conversion dynamics and gonadotropin sensitivity rather than absolute counts alone—and they have demonstrated value for predicting ART outcomes and guiding individualized treatment (Chen et al., 2020; Carosso et al., 2022; Cesarano et al., 2022; Sunkara et al., 2025).

To date, a standardized quantitative framework for defining and characterizing ovarian response to FSH superstimulation in dairy cattle using these KPI remains lacking. We therefore retrospectively analyzed 1,090 FSH-stimulated OPU sessions in Holstein donors to evaluate 3 process-oriented KPI adapted for bovine application: the follicular output rate (**FORT_bov**), the follicle-to-oocyte index (**FOI_bov**), and the ovarian sensitivity index (**OSI_bov**), and to quantify their associations with oocyte yield and quality (including counts and proportions of grade A+B COC); and, using a 3-tier reference classification (poor, normal, high), to compare diagnostic performance for distinguishing poor versus normal + high, and high versus poor + normal responders. Our goal is to provide a practical, quantitative framework for defining OS response in cattle and to inform individualized, efficient OPU-IVP programs.

MATERIALS AND METHODS

Animals and Ethics

This retrospective cohort study analyzed OPU records collected between October 2023 and June 2024 from a commercial dairy farm in Northwest China (37°48′37″N,

Table 2. Baseline characteristics and OPU-related outcomes in dairy cows classified as poor, normal, or high ovarian responders[1]

Item	Poor	Normal	High
Baseline AFC[2]	12.44 ± 6.08^{a}	16.83 ± 6.81^{b}	24.47 ± 8.11^{c}
Total FSH dose	742.74 ± 132.66^{c}	701.31 ± 138.98^{b}	675.37 ± 149.09^{a}
Aspirated follicles (n)	10.68 ± 4.49^{a}	23.82 ± 6.94^{b}	44.97 ± 9.04^{c}
Total COC recovered[3] (n)	5.02 ± 1.92^{a}	12.51 ± 3.93^{b}	25.03 ± 7.22^{c}
COC recovery[4] (%)	50.52 ± 19.60^{a}	54.88 ± 17.59^{b}	56.63 ± 15.63^{b}
Grade A COC (n)	0.63 ± 0.83^{a}	3.04 ± 2.54^{b}	6.70 ± 4.32^{c}
Grade B COC (n)	1.21 ± 1.13^{a}	3.77 ± 2.56^{b}	8.28 ± 4.64^{c}
Grade C COC (n)	1.95 ± 1.19^{a}	3.63 ± 1.75^{b}	5.87 ± 2.93^{c}
Grade D COC (n)	1.23 ± 0.99^{a}	2.07 ± 1.62^{b}	4.18 ± 3.87^{c}
Grade A COC (%)	13.62 ± 19.54^{a}	23.69 ± 17.95^{b}	26.24 ± 14.37^{c}
Grade B COC (%)	23.25 ± 20.42^{a}	29.16 ± 16.05^{b}	32.76 ± 15.05^{c}
Grade C COC (%)	38.50 ± 22.23^{c}	30.19 ± 12.88^{b}	24.32 ± 11.39^{a}
Grade D COC (%)	24.63 ± 15.73^{b}	16.97 ± 11.93^{a}	16.78 ± 12.91^{a}
Grade A+B COC[5] (n)	1.84 ± 1.30^{a}	6.81 ± 3.64^{b}	14.97 ± 6.60^{c}
Grade A+B COC[6] (%)	36.87 ± 21.81^{a}	52.84 ± 18.40^{b}	59.00 ± 17.60^{c}

$^{a–c}$Within a row, means with different superscript letters differ ($P < 0.05$).

[1]Values are means ± SD. Because OPU sessions were repeatedly measured within cows, group differences were tested using models with cow-cluster robust SE (clustered by cow), with multiplicity-adjusted pairwise comparisons when appropriate.

[2]Number of 2- to 8-mm follicles counted before stimulation (random day of the estrous cycle).

[3]Number of follicles actually aspirated during the OPU procedure.

[4]Number of total COC recovered/number of aspirated follicles.

[5]Number of grade A and grade B COC.

[6]Number of grade A+B COC/number of total COC.

106°21′39″E). A total of 1,169 initial OPU records were initially obtained, with all donors being healthy Holstein cows (5–8 yr of age) selected based on genetic merit (parity ≥3 and previous-lactation 305-d milk yield ranking in the top 10% of the herd). Inclusion required complete OPU data, including baseline AFC, number of aspirated follicles, total oocytes recovered, oocyte quality grade (A/B/C/D), and OS protocol. After applying these criteria, 1,090 OPU sessions (corresponding to 481 cows) met eligibility requirements and were included in the final analysis. Donors were maintained under standard commercial conditions and fed a TMR formulated according to routine farm nutritional-management practices. The OPU and stimulation were performed as part of routine reproductive management by the same team (2 veterinarians and 2 technicians—one farm-based and one laboratory-based) to ensure procedural consistency. All procedures complied with relevant animal-welfare guidelines and were approved by the Animal Care and Use Committee of Northwest A&F University.

Ovarian Stimulation

All donor cows underwent a standardized hormonal stimulation protocol before OPU. Ovarian status was assessed via rectal palpation and B-mode ultrasonography (Easi-Scan, IMV Technologies, France) to exclude animals with inactive ovaries, pregnancy, or cystic follicles. On a random day of the estrous cycle, all antral follicles (2–8 mm in diameter) on both ovaries were counted to assess baseline AFC. Follicle ablation (**FA**) was then performed to synchronize the follicular wave. The procedure was carried out as previously described (Hayden et al., 2022), by transvaginal ultrasound-guided aspiration (Exapad-Mini, IMV Technologies, France) of all follicles >5 mm in diameter to synchronize the emergence of a new follicular wave.

Stimulation timelines were as follows. If a corpus luteum (**CL**) was present, 25 mg of $PGF_{2\alpha}$ (Sansheng Biological Technology, Ningbo, China) was administered 24 h after FA to induce luteolysis. Follicle-stimulating hormone (Sansheng Biological Technology, Ningbo, China) treatment was initiated 12 h later and administered twice daily for 2 d, with a total dose of 400 to 1,000 IU per OPU session (equivalent to 200–500 mg, based on the manufacturer's potency information that 100 IU corresponds to 50 mg). Ovum pick-up was performed 36 h after the final FSH injection. If no CL was present, FSH was started 36 h after FA and administered twice daily for 2 d; $PGF_{2\alpha}$ was administered 12 h after the last FSH injection; and OPU followed 24 h later. The total FSH dose administered before each OPU was recorded for analysis.

OPU Procedure

Epidural anesthesia was induced with 5 mL of 2% lidocaine at the tailhead; adequacy was confirmed by

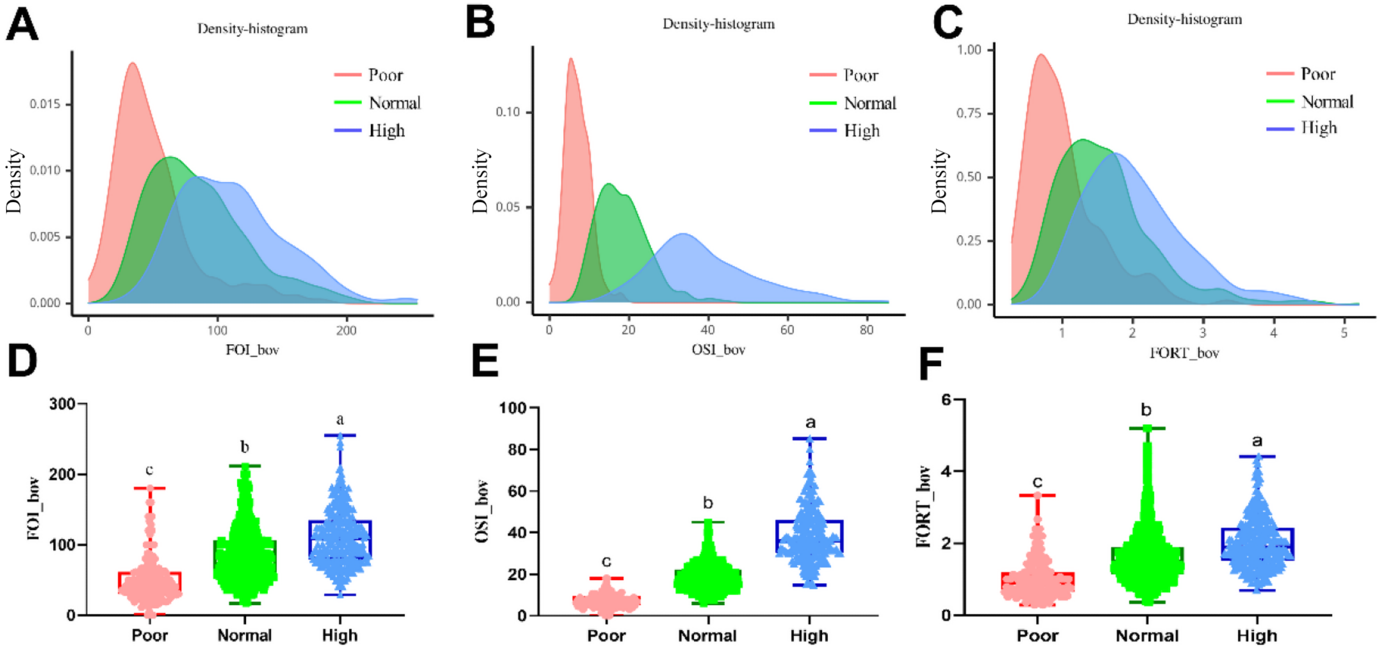

Figure 1. Distribution and group-wise comparison of FOI_bov, OSI_bov, and FORT_bov according to ovarian response category. (A–C) Kernel density plots of FOI_bov (A), OSI_bov (B), and FORT_bov (C) in poor (red), normal (green), and high (blue) responders. (D–F) Boxplots of FOI_bov (D), OSI_bov (E), and FORT_bov (F) for each response group. Horizontal lines indicate medians, boxes the interquartile ranges, and whiskers the minimum and maximum values. Different letters (a–c) above the boxes indicate significant differences among groups ($P < 0.05$) based on models accounting for repeated sessions within cow (cow-cluster robust standard errors), with multiplicity-adjusted pairwise comparisons.

loss of tail tone. After manual evacuation of the rectum and aseptic preparation of the perineum and vulva, a vaginal ultrasound probe was introduced, and the ovaries were positioned rectally to obtain clear images. All follicles >2 mm was aspirated using a disposable OPU needle (1.20 mm × 75 mm, 18-gauge; catalog no. 2207, Misawa Medical Industry Co. Ltd.) connected to a vacuum unit (FV-6; Fuji Ping) calibrated to approximately 70 mm Hg (corresponding to a flow of 20–30 mL/min). Follicles were flushed with prewarmed (38.5°C) Dulbecco's PBS supplemented with 1% fetal calf serum and heparin (stock solution, 5,000 IU/mL). Each follicle was aspirated individually, and fluids from follicles of the same ovary and size class were pooled. At the end of each session, the needle was flushed to recover any COC retained within the lumen. Follicular fluid was transported to a nearby laboratory within 20 min in a temperature-controlled container.

Oocyte Recovery and Grading

The COC were recovered and evaluated under a stereomicroscope (SZX7, Olympus) at 30× magnification. Recovered oocytes were washed 2 to 3 times in PBS and classified morphologically based on cytoplasmic appearance and cumulus cell distribution, with minor modifications from a previously described system (de Loos et al., 1989). Grade A COC exhibited dense, multilayered cumulus investment and homogeneous cytoplasm, with a light and translucent overall appearance. Grade B COC also had compact, multilayered cumulus and uniform cytoplasm but appeared slightly darker and less translucent, often with a darker peripheral band. Grade C COC showed less dense cumulus, irregular cytoplasm with dark clusters, and overall darker coloration. Grade D COC had expanded cumulus with dark cell clumps embedded in a gelatinous matrix, irregular cytoplasm with dark areas, and a generally dark and irregular appearance. In this study, COC with homogeneous cytoplasm, light brown uniform appearance, and at least one layer of compact cumulus cells was considered viable and used for in vitro culture. Partially denuded COC or naked oocytes (grade D) were deemed nonviable and discarded.

Definition of Ovarian Response Categories

To systematically evaluate individual donor responses to superovulation, OPU sessions were categorized by the total counts of aspirated follicles and recovered COC, adopting a method previously described (De Roover et al., 2005). Based on 1,090 eligible OPU records, the overall distribution characteristics of both follicle and COC counts were determined. Response levels were defined using the overall mean ± 1 SD approach. Follicle and

Table 3. Cow-level Spearman correlation coefficients between FOI_bov, OSI_bov, FORT_bov, and ovarian superstimulation outcomes (n = 481 cows)[1]

Item	FOI_bov, r (*P*-value)	OSI_bov, r (*P*-value)	FORT_bov, r (*P*-value)
Baseline AFC	−0.279 ($P < 0.001$)	0.515 ($P < 0.001$)	−0.303 ($P < 0.001$)
Total FSH dose	0.016 ($P = 0.728$)	−0.317 ($P < 0.001$)	0.011 ($P = 0.809$)
Aspirated follicles (n)	0.413 ($P < 0.001$)	0.816 ($P < 0.001$)	0.542 ($P < 0.001$)
Total COC recovered (n)	0.609 ($P < 0.001$)	0.949 ($P < 0.001$)	0.446 ($P < 0.001$)
COC recovery (%)	0.440 ($P < 0.001$)	0.343 ($P < 0.001$)	−0.074 ($P = 0.106$)
Grade A+B COC (n)	0.564 ($P < 0.001$)	0.869 ($P < 0.001$)	0.416 ($P < 0.001$)
Grade A+B COC (%)	0.270 ($P < 0.001$)	0.411 ($P < 0.001$)	0.223 ($P < 0.001$)

[1]Values are Spearman's rank correlation coefficients (r), with *P*-values in parentheses. Because the variables deviated from normality (Shapiro–Wilk test, $P < 0.05$), correlations were assessed using Spearman's rank correlation. Correlations were calculated at the cow level using per-cow mean values across all available OPU sessions to avoid pseudo-replication due to repeated sessions within cow. $P < 0.05$ was considered significant. FOI_bov = follicle-to-oocyte index for bovine application; OSI_bov = ovarian sensitivity index for bovine application; FORT_bov = follicular output rate for bovine application.

COC counts were classified independently according to the same criteria, resulting in a triple-category response system (poor, normal, and high), as summarized in Table 1. This data-driven strategy enables standardized classification of clinical ovarian response, independent of absolute yield, and reduces misclassification bias from extreme values in either variable.

Calculation of FORT, FOI, and OSI

To quantitatively assess ovarian response to FSH stimulation, 3 KPI widely used in human ART were adapted for bovine OPU-IVP systems and designated as FORT_bov, FOI_bov, and OSI_bov:

$$\text{FORT\_bov} = (\text{Aspirated follicles}/\text{Baseline AFC}) \times 100\%$$

$$\text{FOI\_bov} = \text{Total COC recovered}/\text{Baseline AFC}$$

$$\text{OSI\_bov} = \text{Total COC recovered}/\text{Total FSH dose}$$

(results may be multiplied by 1,000 for presentation).

Baseline AFC was the 2- to 8-mm follicle count obtained before stimulation; Aspirated follicles was the number of follicles actually aspirated during the OPU procedure; and the total FSH dose was the cumulative IU administered in the stimulation course preceding that OPU.

Statistical Analysis

All statistical analyses were performed using SAS 9.4 (SAS Institute Inc., Cary, NC). Graphs and receiver operating characteristic (**ROC**) curves were generated with GraphPad Prism 8.0 (GraphPad Software Inc., La Jolla, CA). Analyses were conducted at the session level (OPU session as the observational unit), and within-cow correlation from repeated sessions was addressed using cow-cluster robust methods (clustered by cow). Unless otherwise stated, all tests were 2-sided and $P < 0.05$ was considered significant. Data are presented as mean ± SD, and between-group differences in tables are indicated by different superscript letters (a, b, c).

Continuous variables were assessed for normality (Shapiro–Wilk test) and homogeneity of variance (Levene's test). Because most variables deviated from normality (Shapiro–Wilk test, $P < 0.05$), group differences among poor, normal, and high responders were tested using linear models with cow-cluster robust standard errors, with multiplicity-adjusted pairwise comparisons when appropriate. Spearman correlations were calculated at the cow level (n = 481 cows) using per-cow mean values across sessions to avoid pseudo-replication, and were used to evaluate associations between FOI_bov, OSI_bov, and FORT_bov and major OPU outcomes. Correlation strength was interpreted as very weak (<0.20), weak (0.20–0.39), moderate (0.40–0.59), strong (0.60–0.79), or very strong (≥0.80). The ability of FOI_bov, OSI_bov, and FORT_bov to discriminate low responders (poor vs. normal + high) and high responders (high vs. poor + normal) was assessed by ROC analysis. The area under the ROC curve (**AUC**) and 95% CI were reported and classified as excellent (≥0.90), good (0.80–0.90), fair (0.70–0.80) or poor (<0.70). The AUC 95% CI were obtained using cow-cluster bootstrap resampling (B = 2,000), resampling cows with replacement and retaining all sessions within each resampled cow. Optimal cut-offs were determined using Youden's J statistic, and sensitivity and specificity were reported. Agreement between response categories defined by the optimal cut-off of the best-performing index and the clinical classification based on aspirated follicle number and total oocytes recovered was evaluated using quadratic weighted kappa (κ_w; $\kappa \geq 0.75$, excellent; $0.40 \leq \kappa < 0.75$, moderate; $\kappa < 0.40$, poor).

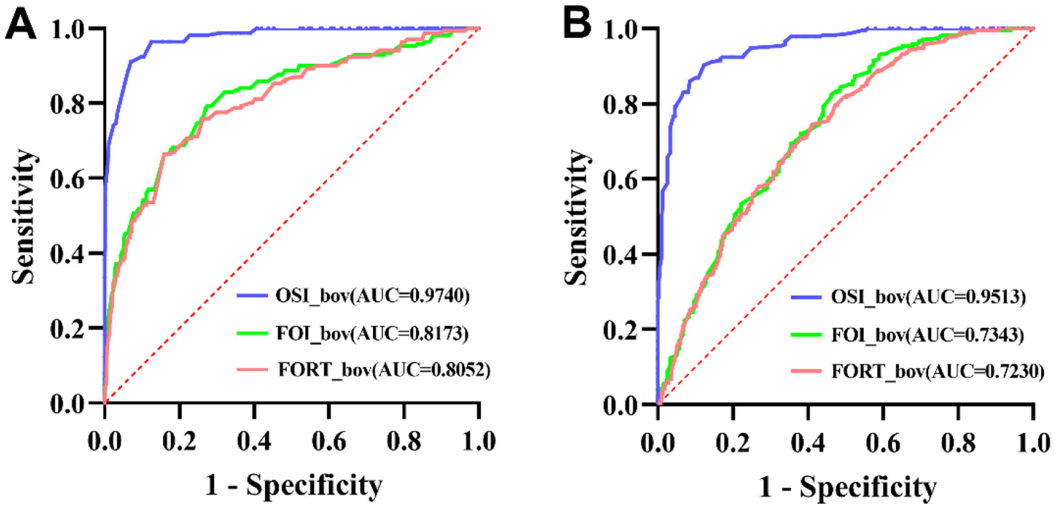

Figure 2. The ROC curves of FOI_bov, OSI_bov and FORT_bov for discriminating ovarian response categories. (A) Poor responders versus normal + high responders. (B) High responders versus poor + normal responders. Blue, OSI_bov; green, FOI_bov; red, FORT_bov. The AUC point estimates are shown in the legend; 95% CI were obtained using cow-cluster bootstrap resampling (B = 2,000) and are reported in Table 4.

To further quantify the discriminatory ability of the 3 indices, ordinal logistic regression models were fitted with ovarian response category (poor < normal < high) as the dependent variable, with FOI_bov, OSI_bov or FORT_bov included as the sole predictor in separate models. Models were fitted with cow-cluster robust SE (clustered by cow). Odds ratios (**OR**) and 95% CI were derived by exponentiating the slope coefficient (β) and correspond to a 1-unit increase in each index. Model performance was compared using the Akaike information criterion (**AIC**) and a generalized concordance index (**c-index**). Within-cow repeatability and session-to-session variability were evaluated among cows with ≥2 OPU sessions. Repeatability was assessed using the intraclass correlation coefficient (**ICC**; 1,1) from a one-way random-effects model, with 95% CI obtained by cow-cluster bootstrap resampling (B = 2,000). Within-cow variability was summarized using the CV (%), calculated for each cow as SD/mean × 100% across repeated sessions and reported as median (quartiles 1–3 [**Q1–Q3**]). The ICC values were interpreted as poor (<0.50), moderate (0.50–0.75), good (0.75–0.90), and excellent (>0.90) repeatability.

RESULTS

Baseline Characteristics and Clinical Stratification of Ovarian Response

A total of 1,090 superstimulation and OPU sessions were obtained from 481 clinically healthy Holstein cows. Of these, 289 cows had ≥2 OPU sessions (898 sessions). Because multiple OPU sessions were available per cow, all inferential analyses accounted for within-cow clustering, and cow-level repeatability/variability was evaluated in cows with repeated sessions. Overall, the mean number of aspirated follicles per session was 26.35 ± 12.98, and the mean number of recovered COC was 14.05 ± 7.86. For the clinical classification of ovarian response, sessions were categorized into 3 groups (poor, normal, and high responders) based on the joint distribution of follicle number and total COC, using an "overall mean ± 1 SD" strategy. According to this classification, poor responders accounted for 170 sessions (15.60%), normal responders for 684 sessions (62.75%), and high responders for 236 sessions (21.65%) of the 1,090 stimulation sessions.

Descriptive characteristics and OPU outcomes according to ovarian response category are shown in Table 2. Baseline AFC, the number of aspirated follicles and total COC recovered increased progressively from poor to normal to high responders, whereas total FSH dose was modestly higher in poor responders and lower in normal and high responders. High responders produced the greatest numbers and proportions of grade A, B, and A+B COC, whereas the proportion of grade C COC declined across the poor-normal-high gradient. The COC recovery rate and the proportion of grade D COC also differed among groups, with generally more favorable oocyte quality profiles in the normal and high responder groups compared with poor responders.

Distribution and Correlations of Ovarian Response Indices with OPU Outcomes

After establishing the clinical ovarian response groups, we next examined how FOI_bov, OSI_bov, and FORT_bov were distributed across the poor, normal,

Table 4. Diagnostic performance of FOI_bov, OSI_bov and FORT_bov for discriminating ovarian response categories based on ROC analysis[1]

Item	AUC (95% CI)	Cut-off	Sensitivity (%)	Specificity (%)
Poor vs. normal + high				
FOI_bov	0.817 (0.769–0.864)	65.00	72.50	79.41
OSI_bov	0.974 (0.962–0.983)	11.00	92.93	91.18
FORT_bov	0.805 (0.748–0.857)	1.03	84.02	66.47
High vs. poor + normal				
FOI_bov	0.734 (0.693–0.773)	73.91	82.63	53.63
OSI_bov	0.951 (0.934–0.966)	25.00	90.25	87.70
FORT_bov	0.723 (0.681–0.767)	1.52	74.58	58.78

[1]Poor versus normal + high, and high versus poor + normal comparisons are shown separately. Values are area under the ROC curve (AUC) with 95% CI in parentheses. Confidence intervals were estimated using cow-cluster bootstrap resampling (B = 2,000). Cut-off values were determined using Youden's J statistic; sensitivity and specificity are reported for each optimal cut-off. FOI_bov = follicle-to-oocyte index for bovine application; OSI_bov = ovarian sensitivity index for bovine application; FORT_bov = follicular output rate for bovine application.

and high categories. The distributions of FOI_bov, OSI_bov, and FORT_bov differed clearly among ovarian response categories (Figure 1). Density plots showed a progressive rightward shift of all 3 indices from poor to normal to high responders (Figure 1A–C), indicating higher values with increasing ovarian response. Consistently, boxplots demonstrated that the median FOI_bov, OSI_bov, and FORT_bov values were lowest in the poor group, intermediate in the normal group and highest in the high group, with significant differences among the 3 response categories (Figure 1D–F; $P < 0.05$, cow-cluster robust comparisons).

Spearman analysis showed that all 3 indices were significantly associated with key OPU outcomes (Table 3; cow-level means, n = 481 cows). Among the 3 indices, OSI_bov showed the strongest correlations with the number of aspirated follicles (r = 0.816), total COC recovered (r = 0.949), and the number of grade A+B COC (r = 0.869; all $P < 0.001$), and was also correlated with the proportion of grade A+B COC (r = 0.411, $P < 0.001$). Similarly, FOI_bov showed positive correlations with total COC, grade A+B COC, COC recovery rate, and aspirated follicles (r = 0.413–0.609, all $P < 0.001$), and a negative correlation with AFC (r = −0.279, $P < 0.001$). By contrast, FORT_bov was correlated with aspirated follicles and total COC recovered (r = 0.542 and 0.446, respectively; $P < 0.001$), showed weaker correlations with grade A+B COC and their proportion (r = 0.416 and 0.223; $P < 0.001$), and was negatively correlated with baseline AFC (r = −0.303, $P < 0.001$) while showing no association with COC recovery rate (r = −0.074, $P = 0.106$).

Comparative Diagnostic Performance of FOI_bov, OSI_bov, and FORT_bov

The ROC analysis indicated that all 3 indices could discriminate ovarian response categories, with OSI_bov yielding the highest AUC in both comparisons (Figure 2, Table 4). AUC 95% CI were estimated using cow-cluster bootstrap resampling. For classifying poor responders versus normal + high responders, OSI_bov achieved an AUC of 0.974 (95% CI 0.962–0.983) and an optimal cut-off of 11 (sensitivity 92.93%, specificity 91.18%). Both FOI_bov and FORT_bov also showed good discrimination, but with lower AUC values (0.817 [0.769–0.864] and 0.805 [0.748–0.857], respectively). For classifying high responders versus poor + normal responders, yielded the highest AUC among the other indices (AUC 0.951 [0.934–0.966]; cut-off 25; sensitivity 90.25%; specificity 87.70%), whereas FOI_bov and FORT_bov showed fair discrimination (0.734 [0.693–0.773] and 0.723 [0.681–0.767], respectively).

OSI_bov as a Key Performance Indicator: Agreement and Regression Analyses

Given the superior diagnostic performance of OSI_bov, we further evaluated this index as a primary KPI for defining ovarian response. Agreement between OSI_bov-based response categories and the clinical response groups was high. Using the quadratic weighted kappa (κ_w), the concordance between the 3 OSI_bov categories (poor, normal, high) and the corresponding clinical categories defined by aspirated follicle number plus total COC recovered was κ_w = 0.777, indicating strong agreement (Figure 3). Most sessions fell on the diagonal of the confusion matrix, with 155, 514, and 213 sessions simultaneously classified as poor, normal and high responders, respectively, by both methods. To further address within-cow stability of classification across repeated OPU sessions, switching patterns for clinical and OSI_bov-based group assignments are summarized in Supplemental Table S1 (see Notes).

To quantify the predictive ability of each index, we fitted single-predictor ordinal logistic regression models with ovarian response category (poor < normal < high) as the outcome (Table 5). Models used cow-cluster robust standard errors to account for repeated sessions within cow. All 3 indices were significantly associated with higher response levels. For FORT_bov, each 1-unit increase was associated with a 3.38-fold increase in the odds of belonging to a higher response category (95% CI: 2.53–4.53; $P < 0.001$), whereas FOI_bov showed a more modest effect (OR = 1.03; 95% CI: 1.02–1.03; $P < 0.001$). The OSI_bov had the strongest association, with an OR of 1.42 per unit increase (95% CI: 1.35–1.50; $P < 0.001$), and showed the best model fit and discrimination among the 3 indices (AIC = 881.7; c-index = 0.956).

Repeatability and Within-Cow Variability Across Repeated OPU Sessions

Repeatability and within-cow variability were evaluated to characterize variation across repeated OPU sessions within cow and to account for the non-independence of session-level records. Analyses were restricted to cows with ≥2 OPU sessions (289 cows; 898 sessions) and summarized using ICC (1,1) with cow-cluster bootstrap 95% CI and within-cow CV (median [Q1–Q3]; Table 6). Baseline AFC was repeatable (ICC = 0.677 [0.614–0.727]) with moderate within-cow variation (CV 18.61% [10.63–28.28]). Aspirated follicle number and total COC recovered also showed high repeatability (ICC = 0.822 [0.783–0.851] and 0.749 [0.704–0.789], respectively) and relatively low within-cow CV (16.03% [8.84–24.05] and 22.05% [13.86–32.26]). In contrast, FOI_bov and FORT_bov were less repeatable (ICC = 0.432 [0.357–0.497] and 0.498 [0.409–0.572]) and showed greater within-cow variation (CV 27.79% [15.44–41.86] and 24.67% [15.48–35.58]). Grade A+B COC exhibited substantial session-to-session variability (CV 41.04% [24.05–54.49]) despite moderate repeatability (ICC = 0.626 [0.551–0.687]). Recovery rate and total FSH dose were poorly repeatable (ICC = 0.140 [0.060–0.221] and 0.064 [−0.004–0.134]), consistent with pronounced session effects and dose adjustments. The ICC (1,k) estimates are provided in Supplemental Table S2 (see Notes) to show how reliability improves when cow-level means are based on multiple sessions (e.g., k = 2, 3, and $\bar{k}$), which is relevant for cow-level evaluation using repeated OPU records.

DISCUSSION

To our knowledge, this is the first study to evaluate FORT, FOI, and OSI in a large-scale bovine OPU model and to quantify their capacity to characterize ovarian response. Using data from 481 healthy Holstein donors and 1,090 superstimulation and OPU sessions, we classified each cycle into poor, normal, or high responders based on the joint distribution of aspirated follicles and total COC, applying an "overall mean ± 1 SD" strategy rather than arbitrarily chosen cut-off. This data-driven approach parallels that used in human ART (Huber et al., 2013) and ruminants (De Roover et al., 2005; Durocher et al., 2006), where ovarian response categories are derived from the empirical distribution of response in large, unselected cohorts, thereby providing a statistically grounded framework for stratification. The correspondence between follicle counts and oocytes retrieved is a central hallmark of whether OS has truly mobilized ovarian potential. Human data have suggested that good agreement between AFC and the number of oocytes retrieved indicates an adequate response (Vrontikis et al., 2010; Poulain et al., 2021). In our bovine model, the number of follicles visible at OPU and total COC increased in parallel from poor to normal to high responders, supporting this concept and offering an operational definition of "good" ovarian response in donor cows.

However, as in humans, relying solely on static reserve markers such as AFC or AMH to predict ovarian response has clear limitations (Alviggi et al., 2018). Although these markers correlate with oocyte yield at a popula-

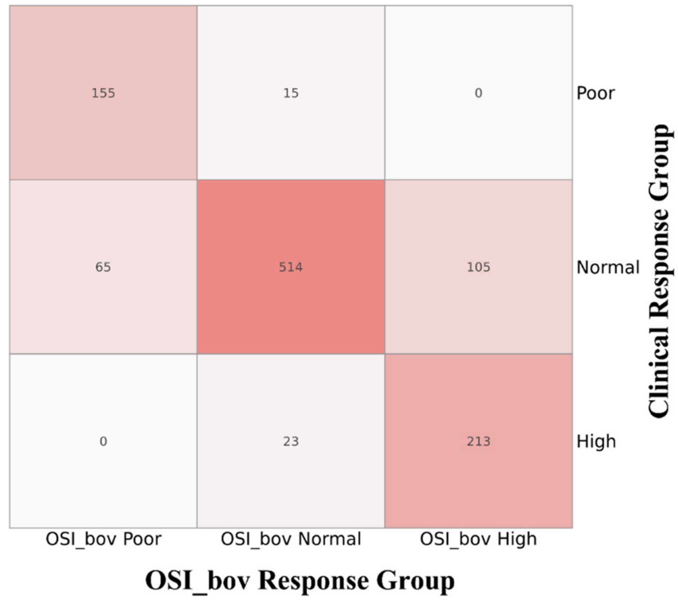

Figure 3. Agreement between OSI_bov-based response categories and clinical ovarian response groups. Heatmap showing the cross-classification of OPU sessions by OSI_bov response group (x-axis: poor, normal, high) and clinical ovarian response group (y-axis: poor, normal, high). Numbers in cells indicate the number of sessions in each classification combination. OSI_bov response groups were defined using ROC-derived cut-offs: poor (OSI_bov <11), normal (11 ≤ OSI_bov <25), and high (OSI_bov ≥25). Clinical response groups were defined from the number of aspirated follicles and total COC recovered. Agreement between the 2 classification schemes was assessed using the quadratic weighted kappa (κ_w).

Table 5. Single-predictor ordinal logistic regression models for ovarian response category (poor < normal < high)[1]

Index	OR (95% CI)	*P*-value	AIC	c-Index
FORT_bov	3.38 (2.53–4.53)	<0.001	1808.6	0.743
FOI_bov	1.03 (1.02–1.03)	<0.001	1766.6	0.754
OSI_bov	1.42 (1.35–1.50)	<0.001	881.7	0.956

[1]Models were fitted separately with follicular output rate for bovine application (FORT_bov), follicle-to-oocyte index for bovine application (FOI_bov), or ovarian sensitivity index for bovine application (OSI_bov) as the sole predictor, using session-level records with cow-cluster robust standard errors (clustered by cow) to account for repeated OPU sessions within cow. Odds ratios (OR) and 95% CI correspond to a 1-unit increase in each index. *P*-values are from Wald tests. AIC = Akaike's information criterion; c-index = generalized concordance index based on pairwise comparisons between predicted scores and observed ordered response levels. $P < 0.05$ was considered significant.

Table 6. Within-cow repeatability and session-to-session variability of ovarian reserve, superstimulation response, and OPU outcomes[1]

Item	ICC (1,1) (95% CI)	CV %, median (Q1–Q3)
Baseline AFC	0.677 (0.614–0.727)	18.61 (10.63–28.28)
Aspirated follicles	0.822 (0.783–0.851)	16.03 (8.84–24.05)
Total COC recovered	0.749 (0.704–0.789)	22.05 (13.86–32.26)
Grade A+B COC	0.626 (0.551–0.687)	41.04 (24.05–54.49)
Recovery rate	0.140 (0.060–0.221)	22.06 (13.56–33.41)
Total FSH dose	0.064 (−0.004–0.134)	16.83 (4.56–24.74)
FOI_bov	0.432 (0.357–0.497)	27.79 (15.44–41.86)
FORT_bov	0.498 (0.409–0.572)	24.67 (15.48–35.58)
OSI_bov	0.711 (0.660–0.753)	25.16 (14.47–36.28)

[1]Values are intraclass correlation coefficients for a single-session measurement [ICC (1,1)] with 95% CI in brackets, and within-cow CV (%) reported as median (Q1–Q3). Analyses were restricted to cows with ≥2 OPU sessions (n = 289 cows; 898 sessions). ICC 95% CI were obtained using cow-cluster bootstrap resampling (B = 2000). CV for each cow was calculated as SD/mean × 100% across that cow's repeated sessions. FOI_bov = follicle-to-oocyte index for bovine application; OSI_bov = ovarian sensitivity index for bovine application; FORT_bov = follicular output rate for bovine application.

tion level, their predictive power for individual cycles is limited (Chen et al., 2020; Roque and Sunkara, 2025). In cattle, several studies have also reported dose-dependent increases in follicle and COC yield with increasing FSH, indicating that stimulation outcomes are not determined by ovarian reserve alone but are strongly influenced by gonadotropin input (Hayden et al., 2023; Motta and Garcia-Guerra, 2025). This underscores the need for dynamic KPI that integrate the entire sequence of "exogenous FSH dose–follicular growth–oocyte yield." Follicle-to-oocyte index, FORT, and OSI were conceived precisely in this context as composite indices capturing different aspects of the stimulation process (Chen et al., 2023; Sunkara et al., 2025).

In our bovine dataset, all 3 indices were significantly associated with key OPU outcomes, but with different strengths. Both FORT_bov and FOI_bov showed moderate correlations with the number of aspirated follicles, total COC, and the number of grade A+B COC, indicating that they reflect the extent to which ovarian reserve is mobilized and utilized. As an index of the proportion of baseline AFC that reach preovulatory or aspiratable size, FORT_bov was clearly lower in poor responders and higher in normal and high responders, in line with human reports where FORT values around 30% indicate under-response and values around 80% indicate adequate response (Alviggi et al., 2018). An intriguing feature in our data was that FORT_bov exceeded 100% in some sessions. Although this appears to contradict the intuitive upper limit of a proportion, it likely reflects the specific follicular dynamics and ultrasound-based counting in cows rather than a true increase beyond the initial follicle pool. The bovine ovary contains a large number of very small antral follicles (2–3 mm) that are difficult to capture in baseline AFC but can rapidly grow to ≥5 to 6 mm under FSH stimulation and thus become visible and countable at OPU (Driancourt, 2001; Ongaratto et al., 2020). Dose-response studies in cattle have also shown that the number of follicles visible at OPU increases almost linearly with FSH dose, partly because larger follicles are easier to visualize and not necessarily because new follicles have been recruited (García Guerra et al., 2015; Hayden et al., 2023). In addition, FA and wave-synchronization protocols can induce the emergence of a new follicular wave before OPU, further increasing the number of visible follicles at a given time point (Burns et al., 2005; Hayden et al., 2022). For these reasons, we interpret FORT_bov >100% as indicating "over-recruitment of visible antral follicles" rather than a literal proportion of the baseline AFC, and suggest that translation of the FORT concept to ruminants should carefully reconsider both the definition and diameter range of baseline AFC and the classification of aspirated follicles at OPU.

FOI_bov, in contrast, condenses the entire OS process into a single ratio of total COC to baseline AFC, serving as an indirect quantitative marker of ovarian responsiveness. In human studies, an FOI ≤50% is often regarded as indicative of low follicular-oocyte output and suboptimal utilization of ovarian potential (Alviggi et al., 2018). However, this threshold remains largely arbitrary, derived predominantly from small, single-center cohorts, and its generalizability across diverse populations and stimulation protocols has not been established (Genro et al., 2012; Sunkara et al., 2025). Furthermore, FOI is influenced not only by ovarian sensitivity to FSH but also considerably by the trigger protocol and oocyte retrieval technique (Wang et al., 2022). Variability in aspiration strategy, operator skill, or equipment may alter the recovered oocyte count, thereby obscuring the underlying biological signal. In our study, FOI_bov showed moderate correlations with the number and proportion of grade A+B COC, but its discriminative performance in ROC analysis and ordinal logistic regression was consistently inferior to that of OSI_bov. These findings suggest that, in the bovine OPU context, FOI_bov may be more appropriate as an auxiliary indicator to detect potential losses in the oocyte recovery process rather than as a standalone metric for defining response categories.

In contrast, OSI_bov demonstrated the most prominent overall advantages in this study. Previous human studies have established OSI as a direct measure of ovarian sensitivity to exogenous gonadotropins, showing high consistency across repeated sessions and outperforming simple oocyte yield in predicting clinical pregnancy, cumulative live birth rate, and embryological success (Revelli et al., 2020; Weghofer et al., 2020; Hsu et al., 2024). Research involving a large, unselected cohort of 7,520 in vitro fertilization or intracytoplasmic sperm injection sessions demonstrated that OSI follows a log-normal distribution, allowing natural stratification into poor, normal, and high responders using mean ± 1 SD, with significant differences in outcomes such as live birth rate (Huber et al., 2013). Our findings in cattle are consistent with this pattern. The OSI_bov showed the strongest correlations with the numbers of aspirated follicles, total COC, and A+B grade oocytes. It demonstrated exceptional diagnostic performance, with AUC of 0.974 for distinguishing poor from normal + high responders and 0.951 for high versus poor + normal responders. In single-indicator ordinal logistic regression, OSI_bov achieved the lowest AIC and highest c-index (0.956), outperforming FOI_bov and FORT_bov in ranking sessions along the poor-normal-high response gradient. Notably, the ROC-derived OSI_bov categories showed strong agreement with the clinical response groups based on follicle and COC counts ($\kappa\_w = 0.777$), indicating that OSI_bov captures biological signals essentially similar to traditional clinical stratification but in a more concise and analytically convenient form.

Because genetically valuable donor cows typically undergo multiple OPU sessions, evaluating within-cow repeatability and session-to-session variability is critical for donor selection and long-term management. In our dataset, baseline AFC and the number of aspirated follicles were highly repeatable across repeated OPU sessions, consistent with previous reports showing that follicle numbers are highly repeatable within individuals across follicular waves and that peak/mean follicle counts across waves can remain relatively stable (Singh et al., 2004; Burns et al., 2005; Ireland et al., 2007). This supports the use of baseline AFC as a relatively stable reference trait when interpreting efficiency indices such as FOI_bov and FORT_bov. Total COC recovered showed moderate repeatability but still exhibited meaningful within-cow variation (median CV = 22%). This is consistent with previous reports indicating that follicle and oocyte yield can be relatively stable over the short term and may increase slightly with repeated induction of follicular waves (Garcia and Salaheddine, 1998), whereas longer-term OPU series suggest that COC yield may decline over time in high-yield donors but remains comparatively stable in low-yield cows (Monteiro et al., 2017; Gamarra-Lacaze et al., 2026). In contrast, traits reflecting oocyte quality and procedural efficiency varied substantially across sessions: grade A+B COC fluctuated markedly, and recovery rate was poorly repeatable. This is biologically plausible because oocyte competence and recoverability depend on the follicular microenvironment and the stage and size distribution of available follicles, and are also influenced by session-specific procedural factors during OPU and COC assessment (Salek et al., 2025). Collectively, these patterns indicate that each OPU session reflects a unique combination of biological and management conditions, underscoring the need for quantitative KPI that can define and monitor stimulation efficiency at the session level rather than relying solely on absolute counts.

From a practical viewpoint, OSI_bov offers several advantages as a primary KPI for OS in donor cows. First, by directly linking gonadotropin input to oocyte output, it aligns closely with the concept of biological cost-effectiveness, which is highly relevant for optimizing gonadotropin dosing in commercial IVP programs. Second, unlike FORT_bov, OSI_bov does not depend on ultrasound-based follicle counts and is therefore less sensitive to variation in image quality, measurement protocols, and operator experience, improving comparability across herds and centers. Third, because donor cows typically undergo multiple OPU sessions, a dose-normalized KPI such as OSI_bov can facilitate longitudinal monitoring and may help inform individualized stimulation management across sessions. Nevertheless, OSI_bov also has limitations: It does not directly reflect ovarian reserve or account for LH supplementation or differences among gonadotropin preparations, and extremely high OSI values may indicate overly conservative FSH dosing with suboptimal oocyte yield rather than truly superior stimulation efficiency. These aspects warrant further investigation in future studies.

CONCLUSIONS

In conclusion, accurate prediction of ovarian response is essential for selecting appropriate stimulation protocols, maximizing oocyte yield, and avoiding under- or over-stimulation in donor cows. In this study, FORT_bov and FOI_bov provided useful information on follicular recruitment and oocyte recovery efficiency, respectively, but OSI_bov emerged as the most reliable and comprehensive indicator of ovarian-stimulation response.

OSI_bov outperformed the other indices in classifying cows into poor, normal, and high responders and showed strong agreement with clinically defined response categories. We therefore propose OSI_bov as a primary KPI to refine OPU protocols and enhance the overall efficiency of embryo production programs in the dairy industry.

NOTES

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Nonstandard abbreviations used: AFC = antral follicle count; AIC = Akaike information criterion; AMH = anti-Müllerian hormone; ART = assisted reproduction; AUC = area under the ROC curve; c-index = concordance index; CL = corpus luteum; COC = cumulus–oocyte complex; ET = embryo transfer; FA = follicle ablation; FOI = follicle-to-oocyte index; FOI_bov = FOI for bovine application; FORT = follicular output rate; FORT_bov = FORT for bovine application; FSH = follicle-stimulating hormone; ICC = intraclass correlation coefficient; IVF = in vitro fertilization; IVP = in vitro embryo production; KPI = key performance indicator; OPU = ovum pick-up; OR = odds ratio; OS = ovarian-stimulation; OSI = ovarian sensitivity index; OSI_bov = OSI for bovine application; Q1–Q3 = quartiles 1–3; ROC = receiver operating characteristic.

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